

SMART EMERGENCY SOUND RECOGNITION WEARABLE FOR THE HEARING IMPAIRED

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Vision of the Department

- Creating ethical leaders in the domain of Artificial intelligence and data Science through effectual teaching and learning process to develop emerging technology solutions for the benefits of industry and society with a focus on holistic learning and excellence

Mission of the Department

- Strengthening basic competencies in the domains of Artificial Intelligence and Data Science Providing high-quality, value-based technical education and developing technology professionals with creative ideas and compelling leadership abilities.
- Using logical thinking to create and develop cutting-edge products in collaboration with industry stakeholders in order to meet global expectations and requirements enabling graduates to adapt to new technologies via strong fundamentals and lifetime learning

Programme Educational Objectives (PEOs)

- Graduates will demonstrate comprehensive expertise in artificial intelligence, machine learning, and data analysis while applying interdisciplinary knowledge in computer science, statistics, and advanced mathematics to address real-world challenges.
- Graduates will innovate and create impactful solutions ethically, contributing to societal betterment, leveraging their leadership, collaboration, and communication skills for effective teamwork and multidisciplinary endeavors.
- Continuously fostering a culture of lifelong learning, graduates will adapt to evolving technological landscapes, demonstrating adaptability and a commitment to ongoing professional development in the dynamic field of AI and data .

Programme Outcomes

Engineering Graduates will be able to:

- **Engineering knowledge:** : Apply knowledge of mathematics, natural science, computing, engineering fundamentals and an engineering specialization as specified in WK1 to WK4 respectively to develop to the solution of complex engineering problems. .
- **Problem analysis:** Identify, formulate, review research literature and analyze complex engineering problems reaching substantiated conclusions with consideration for sustainable development. (WK1 to WK4).
- **Design/development of solutions:** Design creative solutions for complex engineering problems and design/develop systems/components/processes to meet identified needs with consideration for the public health and safety, whole-life cost, net zero carbon, culture, society and environment as required. (WK5)

- **Conduct investigations of complex problems:** Conduct investigations of complex engineering problems using research-based knowledge including design of experiments, modelling, analysis interpretation of data to provide valid conclusions. (WK8)
- **Engineering Tool Usage:** : Create, select and apply appropriate techniques, resources and modern engineering IT tools, including prediction and modelling recognizing their limitations to solve complex engineering problems. (WK2 and WK6).
- **The Engineer and The World:** Analyze and evaluate societal and environmental aspects while solving complex engineering problems for its impact on sustainability with reference to economy, health, safety, legal framework, culture and environment. (WK1, WK5, and WK7)

- **Ethics:** Apply ethical principles and commit to professional ethics, human values, diversity and inclusion; adhere to national international laws. (WK9)
- **Individual and Collaborative Team work:** Function effectively as an individual, and as a member or leader in diverse/multi-disciplinary teams.
- **Communication:** Communicate effectively and inclusively within the engineering community and society at large, such as being able to comprehend and write effective reports and design documentation, make effective presentations considering cultural, language, and learning differences.

- **Project Management and Finance:** Apply knowledge and understanding of engineering management principles and economic decision-making and apply these to one's own work, as a member and leader in a team, and to manage projects and in multidisciplinary environments.
- **Life-Long Learning:** Recognize the need for, and have the preparation and ability for i) independent and life-long learning ii) adaptability to new and emerging technologies and iii) critical thinking in the broadest context of technological change. (WK8)

Programme Specific Outcomes (PSOs)

On the completion of Artificial Intelligence and Data Science program :

- Graduates will be able to apply core principles of science, engineering, and mathematics to develop AI-driven solutions for real-world challenges
- Graduates will design, develop and implement efficient as well as scalable algorithms using programming languages, data structures and cloud computing technologies
- Graduates will integrate ethical, social, and security considerations while developing AI-driven intelligent systems, ensuring responsible AI deployment

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- Smart Emergency Sound Recognition Wearable For The Hearing Impaired is a smart device that uses artificial intelligence to detect and identify important environmental sounds.
- It alerts the user through vibrations and visual messages displayed on a small screen, ensuring they are aware of critical sounds in real time.
- The system improves safety and independence for hearing-impaired individuals by enabling quick recognition of surrounding audio cues.

Problem Statement

People with hearing impairments may miss critical sounds like sirens or alarms, leading to safety risks and reduced independence. A smart wearable device is needed to detect such sounds in real time and provide instant vibration and visual alerts for safety and awareness.

Objectives

- To develop an AI-based model capable of detecting and classifying critical environmental sounds with high accuracy.
- To integrate the classified sounds with a microcontroller system that provides alerts through vibration feedback and visual text display.
- To provide continuous real-time sound awareness that improves personal safety, minimizes reaction delays, and empowers individuals with hearing impairments to live more independently.

Literature Review

Paper Number	Paper Type and Name of publisher	Title, Year, Author	Inferences
One	Article, MDPI	Real-Time Detection of Important Sounds with a Wearable Vibration Based Device for Hearing-Impaired People, 2018, Mete Yağanoğlu, Cermal Köse	This paper presents an experimental and simulation-based approach to real-time sound detection using a wearable device that converts critical environmental sounds into vibration alerts for hearing-impaired users. The authors used machine learning-based classification techniques implemented through Python and embedded platforms like Arduino. The system successfully identified emergency sounds such as sirens etc..

Literature Review

No.	Paper Type and Publisher	Title, Year, Authors	Inferences
Two	Conference Paper, IEEE	SmartBelt: A Wearable Microphone Array for Sound Source Localization with Haptic Feedback, 2022, Simon Michaud et al.	Introduces SmartBelt, a wearable device with microphone arrays that localize sound and provide directional haptic feedback. Achieved 2.9° localization error and 92.3% correct haptic motor activation. Useful for hearing-impaired users but limited by hardware complexity and performance under low-frequency sounds.

Literature Review

Paper Number	Paper Type and Name of publisher	Title, Year, Author	Inferences
Three	Conference Paper, IEEE Xplore	A Deep Learning Based Wearable Healthcare IoT Device for AI-Enabled Hearing Assistance Automation, 2020, F. YOUNG, L. ZHANG, R. JIANG, H. LIU and C. WALL	The literature presents a deep learning-powered wearable IoT device for hearing-impaired individuals, which converts speech to text using Google's API and displays it on a micro-screen for conversation assistance. It also uses an Inception-v4 model to detect emergency sounds like horns and fire alarms with 92% accuracy, providing real-time alerts and enhancing user safety and communication.

Gaps identified

- Limited real-time sound classification in existing devices
- Poor adaptability of current systems to diverse and noisy real-world environments.
- Existing wearable systems for the hearing-impaired either lack sound classification capability or rely heavily on internet-based services and complex hardware, making them unsuitable for offline use and low-cost deployment.

Proposed System

- A wearable device that detects important environmental sounds in real time.
- Uses a microphone to continuously capture surrounding audio signals.
- An AI-based sound classification model identifies critical sounds such as sirens, alarms etc.
- A microcontroller processes the output and triggers alerts.
- Alerts are given through:
 - Vibration motor (haptic feedback)
 - OLED/LED display (visual alert)
- Lightweight, portable, and cost-effective.
- Enhances safety, awareness, and independence.

Model Architecture

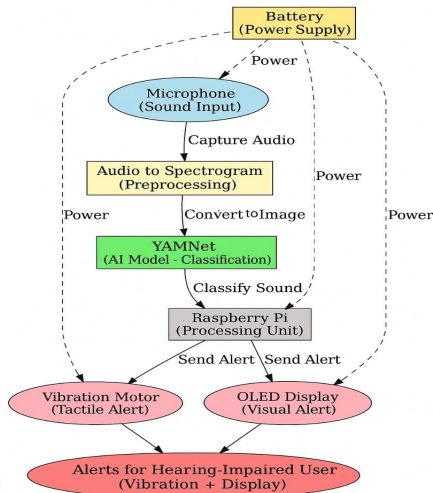


Figure 1: Architecture of proposed method

Architecture WorkFlow

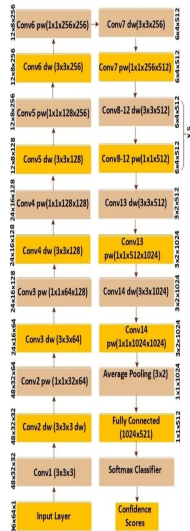


Figure 2: Model Architecture of proposed method

Initial Hardware Arrangement

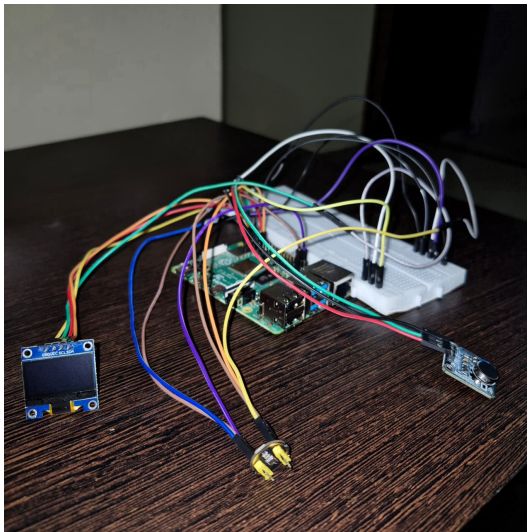


Figure 3: Hardware arrangement of proposed system

Hardware Requirements

- **Raspberry Pi:** Serves as the core processing unit, running the AI model and handling input/output operations.
- **Microphone:** Captures real-time environmental sounds for classification.
- **Vibration Motor:** Provides haptic feedback to notify the user when a critical sound is detected.
- **Display Module:** OLED or small screen used to show visual alerts like sound labels or icons.

Software Requirements

- **Operating System:** Raspberry Pi OS , Windows/Linux for a PC-based system
- **Programming Language:** Python 3.x
- **Libraries and Frameworks:**
 - TensorFlow / PyTorch (for sound classification model)
 - NumPy (for data handling)
 - RPi.GPIO or gpiozero (for motor and display control)
- **Development Environment:** VS Code / Jupyter Notebook

ESC-50 Dataset (Modified):

- A benchmark dataset for environmental sound classification (ESC-50) was used
- A subset of 9 classes was selected from ESC-50 (e.g., siren, dog bark, door knock)
- 4 custom sound classes were created and added to address project-specific needs
- The final dataset contains 13 classes in total (9 from ESC-50 + 4 custom)
- The dataset was used to train the AI model for recognizing and classifying critical sounds

Dataset Used

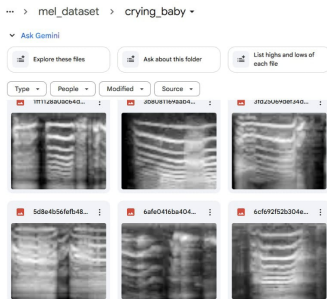


Figure 4: Spectrogram of Crying Baby

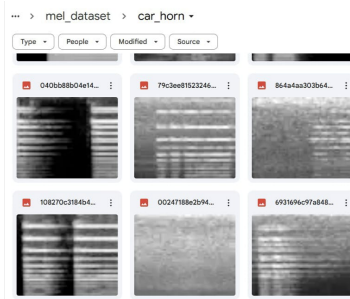


Figure 5: Spectrogram of Car Horn

- Signal Conditioning and Pre-processing

```
# --- STEP 1: SIGNAL REFINEMENT ---  
def prepare_input(self):  
    # Flatten 1-second audio buffer  
    raw_audio = np.concatenate(self.audio_buffer).flatten()  
  
    # 150Hz High-Pass Filter: Removes low-end noise/hum  
    nyquist = 0.5 * SAMPLE_RATE  
    b, a = butter(5, 150 / nyquist, btype='high')  
    return lfilter(b, a, raw_audio).astype(np.float32)
```

Figure 6: Sound Pre-processing

- Feature Extraction and AI Classification

```
# --- STEP 2: HYBRID FEATURE EXTRACTION ---  
def classify_sound(self, clean_audio):  
    # Generate YAMNet Embeddings (Deep Learning Layer)  
    self.interpreter.set_tensor(input_index, clean_audio)  
    self.interpreter.invoke()  
    embeddings = self.interpreter.get_tensor(output_index)  
  
    # Hybrid Strategy: Capture both steady-state and peak features  
    hybrid_emb = np.concatenate([np.mean(embeddings, axis=0),  
                                  np.max(embeddings, axis=0)])  
  
    # Final Classification via SVM  
    probs = svm_model.predict_proba(hybrid_emb.reshape(1, -1))[0]  
    return classes[np.argmax(probs)], np.max(probs)
```

Figure 7: The AI Brain

- Multi-Modal Alert and Hardware Response

```
# --- STEP 3: DYNAMIC ALERTS ---  
def trigger_alert(self, label, confidence):  
    # Display Result on SH1106 OLED  
    with canvas(display) as draw:  
        draw.text((5, 15), "▲ SOUND DETECTED", fill="white")  
        draw.text((5, 35), f"CLASS: {label.upper()}", fill="white")  
  
    # Haptic Feedback: Pulse the Vibration Motor  
    vibration_motor.on()  
    time.sleep(0.5)  
    vibration_motor.off()
```

Figure 8: Response and Feedback

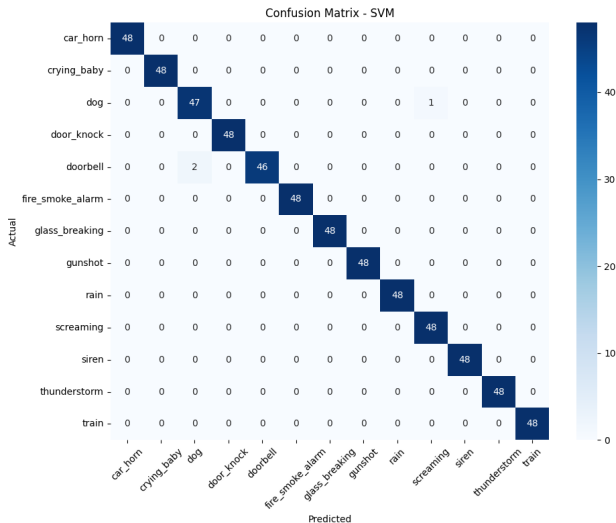


Figure 9: Confusion Matrix

Confusion matrix explanation:

- The confusion matrix shows how well the model classified each sound class.
- Each row represents the actual class, and each column represents the predicted class.
- The diagonal values indicate correctly predicted samples for each class.
- Most classes (like car horn, crying baby, door knock, gunshot, etc.) have all samples correctly classified (value 48).
- A few minor misclassifications are observed (e.g., dog and doorbell show small off-diagonal values).
- This indicates the model performs with very high accuracy and only struggles slightly with similar sound patterns.

Real-Time Audio Classification Output

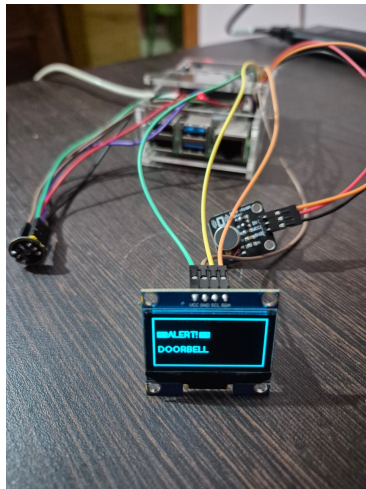
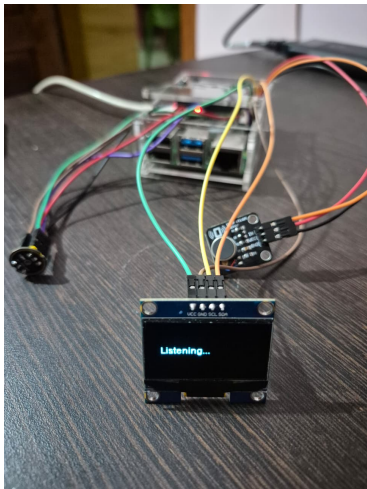


Fig 7: Listening Mode and Doorbell Detection

Real-Time Audio Classification Output

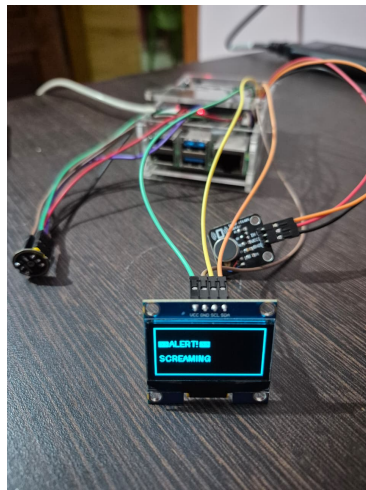
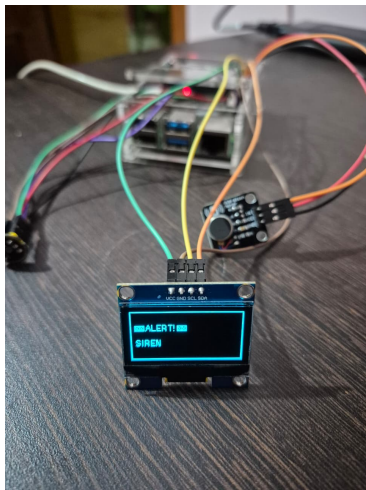


Fig 8: Siren and Screaming Detection

Real-Time Audio Classification Output

Fig 7: Listening Mode and Doorbell Detection

The system initially operates in listening mode, where it continuously captures environmental audio signals. When a doorbell sound is detected, the trained AI model classifies the input in real time and immediately displays the corresponding alert message on the screen.

Fig 8: Siren and Screaming Detection

The system successfully identifies emergency sounds such as siren and screaming. Upon detection, the model performs real-time classification and generates an instant visual alert to notify the user.

Expenses

Item Name	Quantity	Expense (₹)
Raspberry Pi 4 Model B (4GB RAM)	1	5779
Acrylic Case with Fan (Raspberry Pi 4)	1	261
MicroSD Card 64GB (HP A1 V30)	1	650
OLED 1.3 inch Display	1	320
Jumper Wires	30	60
Breadboard	1	70
MEMS High Precision Microphone Module	1	189
DC Vibration Motor Module	1	66
Total Cost	—	7395

Table 2: Summary of Purchased Components and Total Cost

Conclusion

- The proposed wearable system offers real-time sound awareness for the hearing impaired using AI.
- It detects important environmental sounds and alerts users through vibration and visual display.
- By enhancing awareness and safety, this device helps improve daily life independence and responsiveness.
- The system is cost-effective, portable, and easy to use, making it suitable for real-world deployment.

References

- [1] M. Yağanoğlu and C. Köse, “Real-time detection of important sounds with a wearable vibration based device for hearing-impaired people,” *Electronics*, vol. 7, no. 4, p. 50, 2018.
- [2] S. Michaud, B. Moffett, A. Tapia Rousiouk, V. Duda, and F. Grondin, “Smartbelt: A wearable microphone array for sound source localization with haptic feedback,” in *Proceedings of the IEEE International Conference on Robotics and Automation (ICRA)*, IEEE, 2022.
- [3] F. Young, L. Zhang, R. Jiang, H. Liu, and C. Wall, “A deep learning based wearable healthcare iot device for ai-enabled hearing assistance automation,” in *2020 International Conference on Machine Learning and Cybernetics (ICMLC)*, (Adelaide, Australia), IEEE, IEEE, 2020.

THANK YOU